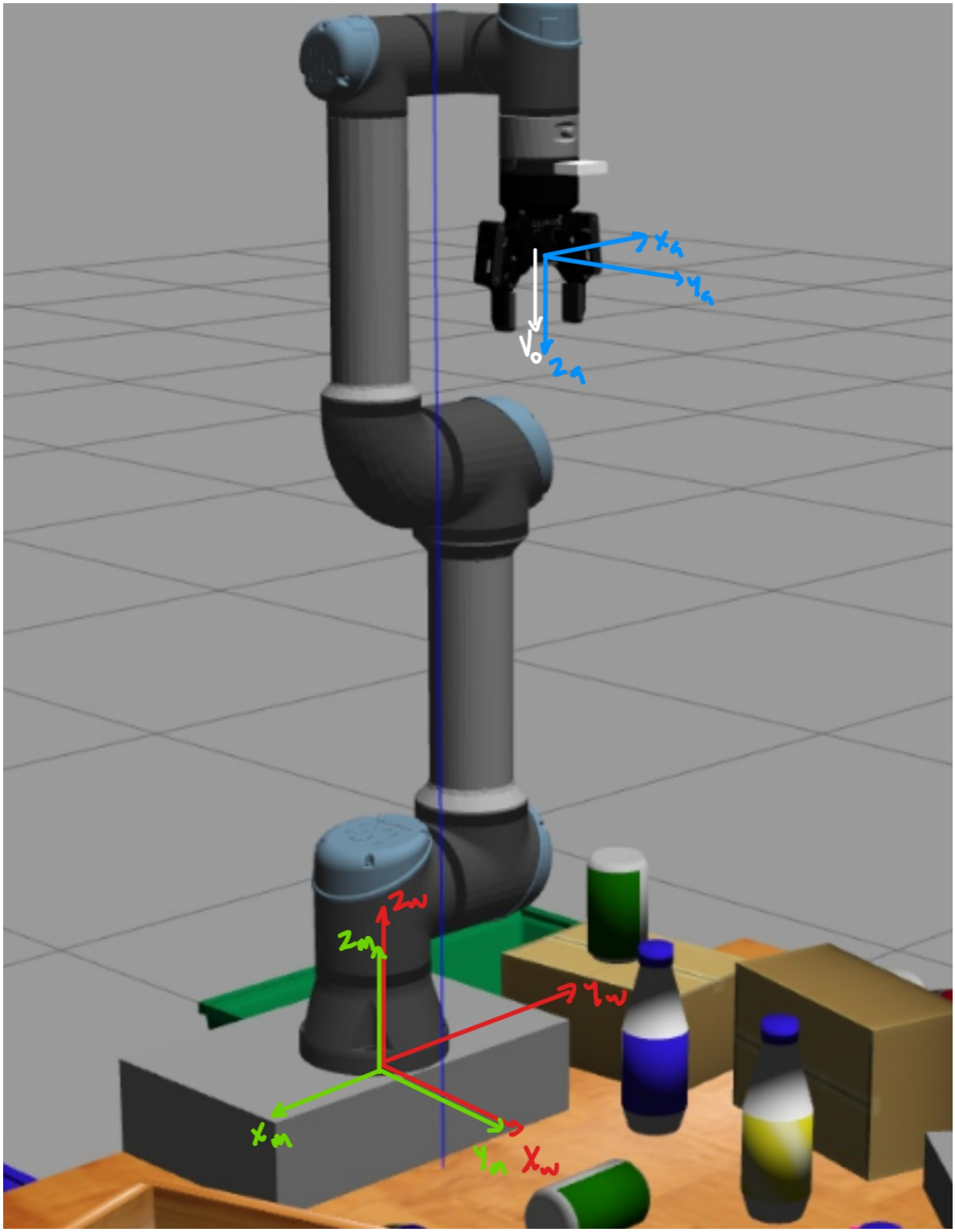
HW 08 – Get Poses

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Coordinate Frame Diagram:



Pseudocode:

get_robot_object_pose_wrt_base_link.m

FUNCTION calculate gripper and model poses wrt the base
    GET robot's handle
    INITIALIZE frame and tool adjustment flags
    GET tool z offset
    GET robot pose wrt to gazebo world coordinate frame
    GET model pose wrt to gazebo wcf
    TRANSFORM robot pose to matlab wcf
    TRANSFORM model pose to matlab wcf
    CALCULATE transformation from robot base to the model
    APPLY model z offset for picking operation
    UPDATE tool orientation for picking operation
    RETURN model position wrt to robot base

    SELECT end effector
    GET end effector pose relative to the robot base
    RETURN end effector position transformed to matlab wcf

ros2matlabPose.m

FUNCTION convert gazebo wcf pose to matlab
    INITIALIZE frame and tool flags
    GET tool offset
    IF no frame adjustment is needed
        IF the message type is gazebo_msgs/GetModelStateResponse
            GET position and orientation from geometry_msgs/Pose
        ELSEIF the message type is geometry_msgs/TransformStamped
            GET position and orientation from geometry_msgs/TransformStamped
    ELSE
        IF the message type is gazebo_msgs/GetModelStateResponse
            GET matlab coordinate frame corrected position and orientation from geometry_msgs/Pose
        ELSEIF the message type is geometry_msgs/TransformStamped
            GET matlab coordinate frame correctedposition and orientation from geometry_msgs/TransformStamped
    RETURN pose as SE3 object

Link:

Code: https://github.com/mrblanders/26F_matlab_ros_support/tree/main/hw/hw08_get_matlab_goal_pose

Poses:

```
Resetting the world...
Getting Gazebo object goal pose(s)...
Picking up model: rCan3
Setting the goal...
    "Base to tip Transform: "

   -0.0028    1.0000   -0.0031    0.1375
    1.0000    0.0027   -0.0043    0.0947
   -0.0043   -0.0031   -1.0000    0.8795
         0         0         0    1.0000

    "Base to Object Transform: "

   -0.0000    1.0000   -0.0000    0.7999
    1.0000    0.0000    0.0000    0.0321
    0.0000         0   -1.0000    0.0482
         0         0         0    1.0000
```